

Администрирование сетевых подсистем

Базовая защита от атак типа «brute force» с использованием Fail2ban

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Цели и задачи работы

Цель лабораторной работы

Получить практические навыки настройки и использования Fail2ban для базовой защиты сервера от атак типа «brute force» на примере служб SSH, HTTP и почтовых сервисов.

Выполнение лабораторной работы

Установка и запуск Fail2ban

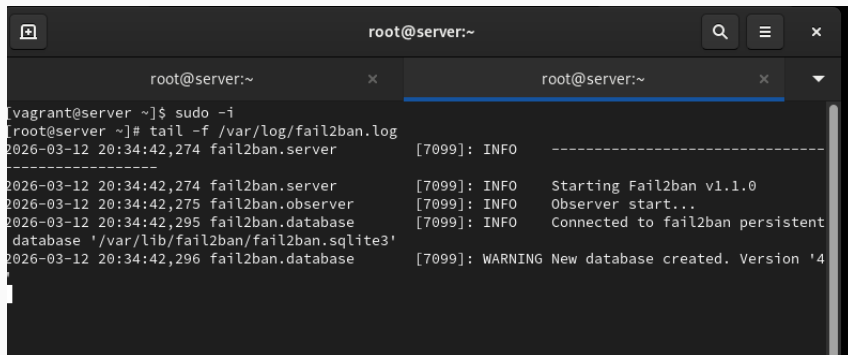
```
Installing      : fail2ban-server-1.1.0-6.el9.noarch      2/5
Running scriptlet: fail2ban-server-1.1.0-6.el9.noarch      2/5
Installing      : fail2ban-firewalld-1.1.0-6.el9.noarch    3/5
Installing      : fail2ban-sendmail-1.1.0-6.el9.noarch    4/5
Installing      : fail2ban-1.1.0-6.el9.noarch              5/5
Running scriptlet: fail2ban-selinux-1.1.0-6.el9.noarch     5/5
Running scriptlet: fail2ban-1.1.0-6.el9.noarch            5/5
Verifying       : fail2ban-1.1.0-6.el9.noarch             1/5
Verifying       : fail2ban-firewalld-1.1.0-6.el9.noarch   2/5
Verifying       : fail2ban-selinux-1.1.0-6.el9.noarch     3/5
Verifying       : fail2ban-sendmail-1.1.0-6.el9.noarch    4/5
Verifying       : fail2ban-server-1.1.0-6.el9.noarch      5/5

Installed:
  fail2ban-1.1.0-6.el9.noarch                fail2ban-firewalld-1.1.0-6.el9.noarch
  fail2ban-selinux-1.1.0-6.el9.noarch         fail2ban-sendmail-1.1.0-6.el9.noarch
  fail2ban-server-1.1.0-6.el9.noarch

Complete!
[root@server ~]# systemctl start fail2ban
[root@server ~]# systemctl enable fail2ban
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /usr/lib/systemd/system/fail2ban.service.
[root@server ~]# S
```

Рис. 1: Просмотр журнала Fail2ban после запуска сервиса

Настройка защиты SSH



A terminal window titled 'root@server:~' with search, menu, and close icons. It shows two tabs, both labeled 'root@server:~'. The active tab displays the following commands and output:

```
[vagrant@server ~]$ sudo -i
[root@server ~]# tail -f /var/log/fail2ban.log
2026-03-12 20:34:42,274 fail2ban.server [7099]: INFO -----
-----
2026-03-12 20:34:42,274 fail2ban.server [7099]: INFO Starting Fail2ban v1.1.0
2026-03-12 20:34:42,275 fail2ban.observer [7099]: INFO Observer start...
2026-03-12 20:34:42,295 fail2ban.database [7099]: INFO Connected to fail2ban persistent
database '/var/lib/fail2ban/fail2ban.sqlite3'
2026-03-12 20:34:42,296 fail2ban.database [7099]: WARNING New database created. Version '4'
```

Рис. 2: Настройка защиты SSH

Проверка запуска jail SSH



```
1 [DEFAULT]
2 bantime = 3600
3
4 #
5 # SSH servers
6 #
7 [sshd]
8 port = ssh,2022
9 enabled = true
10 [sshd-ddos]
11 filter = sshd
12 enabled = true
13 [selinux-ssh]
14 enabled = true
```

Рис. 3: Журнал Fail2ban после включения защиты SSH

Настройка защиты веб-служб (Apache)

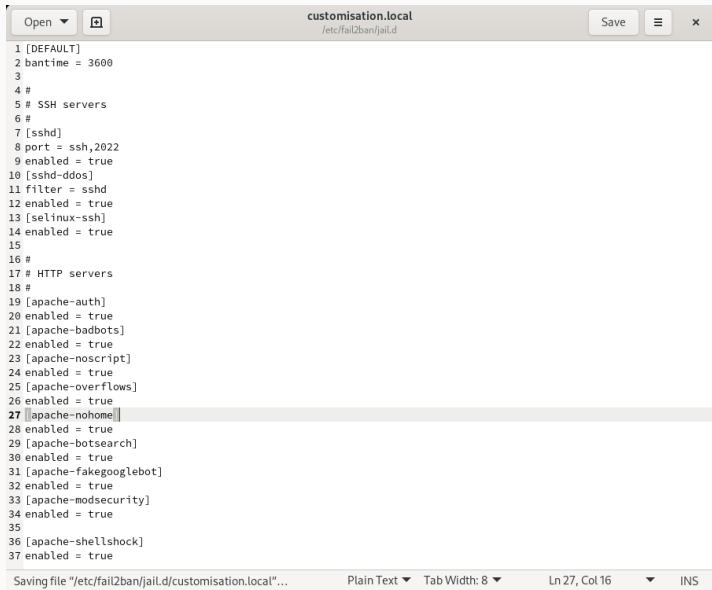
```
(gedit:7201): dconf-WARNING **: 20:37:18.502: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server ~]# systemctl restart fail2ban
[root@server ~]# tail -f /var/log/fail2ban.log
2026-03-12 20:37:31,227 fail2ban.jail      [7230]: INFO      Initiated 'polling' backend
2026-03-12 20:37:31,229 fail2ban.filter  [7230]: INFO      maxLines: 1
2026-03-12 20:37:31,238 fail2ban.filter  [7230]: INFO      maxRetry: 5
2026-03-12 20:37:31,239 fail2ban.filter  [7230]: INFO      findtime: 600
2026-03-12 20:37:31,239 fail2ban.actions [7230]: INFO      banTime: 3600
2026-03-12 20:37:31,239 fail2ban.filter  [7230]: INFO      encoding: UTF-8
2026-03-12 20:37:31,245 fail2ban.filtersystemd [7230]: INFO      [sshd] Jail is in operation now
(process new journal entries)
2026-03-12 20:37:31,246 fail2ban.jail      [7230]: INFO      Jail 'sshd' started
2026-03-12 20:37:31,249 fail2ban.jail      [7230]: INFO      Jail 'selinux-ssh' started
2026-03-12 20:37:31,259 fail2ban.jail      [7230]: INFO      Jail 'sshd-ddos' started
```

Рис. 4: Включение защиты HTTP-сервисов

Проверка защиты HTTP

```
database '/var/lib/fail2ban/fail2ban.sqlite3'
2026-03-12 20:37:31,146 fail2ban.jail [7230]: INFO Creating new jail 'sshd'
2026-03-12 20:37:31,150 fail2ban.jail [7230]: INFO Jail 'sshd' uses systemd {}
2026-03-12 20:37:31,151 fail2ban.jail [7230]: INFO Initiated 'systemd' backend
2026-03-12 20:37:31,156 fail2ban.filter [7230]: INFO maxLines: 1
2026-03-12 20:37:31,180 fail2ban.filterssystemd [7230]: INFO [sshd] Added journal match for:
'_SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session'
2026-03-12 20:37:31,180 fail2ban.filter [7230]: INFO maxRetry: 5
2026-03-12 20:37:31,180 fail2ban.filter [7230]: INFO findtime: 600
2026-03-12 20:37:31,180 fail2ban.actions [7230]: INFO banTime: 3600
2026-03-12 20:37:31,180 fail2ban.filter [7230]: INFO encoding: UTF-8
2026-03-12 20:37:31,181 fail2ban.jail [7230]: INFO Creating new jail 'selinux-ssh'
2026-03-12 20:37:31,182 fail2ban.jail [7230]: INFO Jail 'selinux-ssh' uses poller {}
2026-03-12 20:37:31,182 fail2ban.jail [7230]: INFO Initiated 'polling' backend
2026-03-12 20:37:31,183 fail2ban.datedetector [7230]: INFO date pattern '': 'Epoch'
2026-03-12 20:37:31,184 fail2ban.filter [7230]: INFO maxRetry: 5
2026-03-12 20:37:31,184 fail2ban.filter [7230]: INFO findtime: 600
2026-03-12 20:37:31,184 fail2ban.actions [7230]: INFO banTime: 3600
2026-03-12 20:37:31,184 fail2ban.filter [7230]: INFO encoding: UTF-8
2026-03-12 20:37:31,224 fail2ban.filter [7230]: INFO Added logfile: '/var/log/audit/audit.log' (pos = 0, hash = 0c2be31ffde8214ab616b0e504468ebe00b577f7)
2026-03-12 20:37:31,224 fail2ban.jail [7230]: INFO Creating new jail 'sshd-ddos'
2026-03-12 20:37:31,226 fail2ban.jail [7230]: INFO Jail 'sshd-ddos' uses poller {}
2026-03-12 20:37:31,227 fail2ban.jail [7230]: INFO Initiated 'polling' backend
2026-03-12 20:37:31,229 fail2ban.filter [7230]: INFO maxLines: 1
2026-03-12 20:37:31,238 fail2ban.filter [7230]: INFO maxRetry: 5
2026-03-12 20:37:31,239 fail2ban.filter [7230]: INFO findtime: 600
2026-03-12 20:37:31,239 fail2ban.actions [7230]: INFO banTime: 3600
2026-03-12 20:37:31,239 fail2ban.filter [7230]: INFO encoding: UTF-8
2026-03-12 20:37:31,245 fail2ban.filterssystemd [7230]: INFO [sshd] Jail is in operation now
(process new journal entries)
2026-03-12 20:37:31,246 fail2ban.jail [7230]: INFO Jail 'sshd' started
2026-03-12 20:37:31,249 fail2ban.jail [7230]: INFO Jail 'selinux-ssh' started
2026-03-12 20:37:31,259 fail2ban.jail [7230]: INFO Jail 'sshd-ddos' started
```

Настройка защиты почтовых сервисов



```
1 [DEFAULT]
2 bantime = 3600
3
4 #
5 # SSH servers
6 #
7 [sshd]
8 port = ssh,2022
9 enabled = true
10 [sshd-ddos]
11 filter = sshd
12 enabled = true
13 [selinux-ssh]
14 enabled = true
15
16 #
17 # HTTP servers
18 #
19 [apache-auth]
20 enabled = true
21 [apache-badbots]
22 enabled = true
23 [apache-noscript]
24 enabled = true
25 [apache-overflows]
26 enabled = true
27 [apache-nohome]
28 enabled = true
29 [apache-botsearch]
30 enabled = true
31 [apache-fakegooglebot]
32 enabled = true
33 [apache-modsecurity]
34 enabled = true
35
36 [apache-shellshock]
37 enabled = true
```

Save

customisation.local
/etc/fail2ban/jail.d

Open

Plain Text Tab Width: 8 Ln 27, Col 16 INS

Saving file "/etc/fail2ban/jail.d/customisation.local"...

Проверка защиты почты

```
root@server:~  
root@server:~ x root@server:~ x root@server:~ x  
** (gedit:7296): WARNING **: 20:39:15.499: Set document metadata failed: Setting attribute metadata::gedit-position not supported  
  
(gedit:7296): dconf-WARNING **: 20:39:15.517: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)  
[root@server ~]# systemctl restart fail2ban  
[root@server ~]# tail -f /var/log/fail2ban.log  
2026-03-12 20:39:27,417 fail2ban.filter [7328]: WARNING [apache-botsearch] Please check a jail for a timing issue. Line with odd timestamp: [Thu Mar 12 20:25:24.302442 2026] [ssl:warn] [pid 1369:tid 1369] AH01882: Init: this version of mod_ssl was compiled against a newer library (OpenSSL 3.5.1 1 Jul 2025 (OpenSSL 3.0.7 1 Nov 2022), version currently loaded is 0x30000070) - may result in undefined or erroneous behavior  
2026-03-12 20:39:27,431 fail2ban.jail [7328]: INFO Jail 'apache-botsearch' started  
2026-03-12 20:39:27,440 fail2ban.jail [7328]: INFO Jail 'apache-fakegooglebot' started  
2026-03-12 20:39:27,448 fail2ban.filter [7328]: WARNING [apache-modsecurity] Ignoring all log entries older than 600s; these are probably messages generated while fail2ban was not running.  
2026-03-12 20:39:27,448 fail2ban.filter [7328]: WARNING [apache-modsecurity] Please check a jail for a timing issue. Line with odd timestamp: [Thu Mar 12 20:25:24.302442 2026] [ssl:warn] [pid 1369:tid 1369] AH01882: Init: this version of mod_ssl was compiled against a newer library (OpenSSL 3.5.1 1 Jul 2025 (OpenSSL 3.0.7 1 Nov 2022), version currently loaded is 0x30000070) - may result in undefined or erroneous behavior  
2026-03-12 20:39:27,449 fail2ban.jail [7328]: INFO Jail 'apache-modsecurity' started  
2026-03-12 20:39:27,461 fail2ban.filter [7328]: WARNING [apache-shellshock] Ignoring all log entries older than 600s; these are probably messages generated while fail2ban was not running.  
2026-03-12 20:39:27,470 fail2ban.jail [7328]: INFO Jail 'apache-shellshock' started  
2026-03-12 20:39:27,470 fail2ban.filter [7328]: WARNING [apache-shellshock] Please check a jail for a timing issue. Line with odd timestamp: [Thu Mar 12 20:25:24.302442 2026] [ssl:warn] [pid 1369:tid 1369] AH01882: Init: this version of mod_ssl was compiled against a newer library (OpenSSL 3.5.1 1 Jul 2025 (OpenSSL 3.0.7 1 Nov 2022), version currently loaded is 0x30000070) - may result in undefined or erroneous behavior  
2026-03-12 20:39:27,508 fail2ban.jail [7328]: INFO Jail 'sshd-ddos' started
```

Проверка статуса Fail2ban

```
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:    0
|   `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`-- Actions
    |- Currently banned: 0
    |- Total banned:    0
    `-- Banned IP list:

[root@server ~]# fail2ban-client set sshd maxretry 2
2
[root@server ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:    0
|   `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`-- Actions
    |- Currently banned: 0
    |- Total banned:    0
    `-- Banned IP list:

[root@server ~]# fail2ban-client set sshd unbanip <ip-адрес клиента>
-bash: syntax error near unexpected token `newline'
[root@server ~]# fail2ban-client set sshd unbanip 128.45.29.1.1
0
[root@server ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|   |- Currently failed: 0
|   |- Total failed:    0
|   `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`-- Actions
    |- Currently banned: 0
    |- Total banned:    0
    `-- Banned IP list:

[root@server ~]#
```

Блокировка IP-адреса клиента



Рис. 9: Блокировка IP-адреса клиента

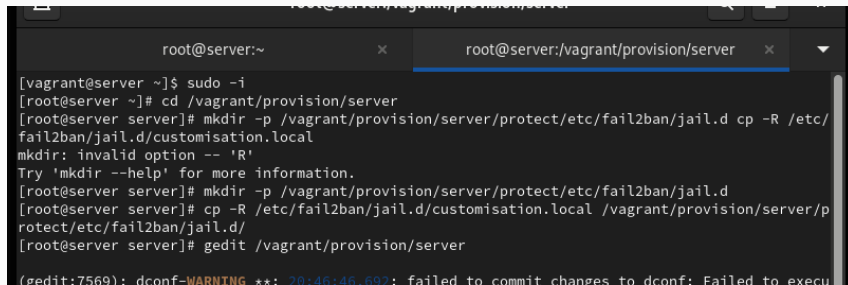
Разблокировка и исключение IP-адреса

```
** (gedit:7438): WARNING **: 20:44:29.558: Set document metadata failed: Setting attribute metadata::gedit-position not supported

(gedit:7438): dconf-WARNING **: 20:44:29.584: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server ~]# systemctl restart fail2ban
[root@server ~]# tail -f /var/log/fail2ban.log
2026-03-12 20:44:38,221 fail2ban.jail [7469]: INFO Jail 'apache-auth' started
2026-03-12 20:44:38,247 fail2ban.jail [7469]: INFO Jail 'apache-badbots' started
2026-03-12 20:44:38,251 fail2ban.jail [7469]: INFO Jail 'apache-noscript' started
2026-03-12 20:44:38,253 fail2ban.jail [7469]: INFO Jail 'apache-overflows' started
2026-03-12 20:44:38,257 fail2ban.jail [7469]: INFO Jail 'apache-nohome' started
2026-03-12 20:44:38,294 fail2ban.jail [7469]: INFO Jail 'apache-botsearch' started
2026-03-12 20:44:38,317 fail2ban.jail [7469]: INFO Jail 'apache-fakegooglebot' started
2026-03-12 20:44:38,360 fail2ban.jail [7469]: INFO Jail 'apache-modsecurity' started
2026-03-12 20:44:38,392 fail2ban.jail [7469]: INFO Jail 'apache-shellshock' started
2026-03-12 20:44:38,402 fail2ban.jail [7469]: INFO Jail 'sshd-ddos' started
```

Рис. 10: Настройка ignoreip

Проверка ignoreip



```
root@server:~  
root@server:/vagrant/provision/server  
[vagrant@server ~]$ sudo -i  
[root@server ~]# cd /vagrant/provision/server  
[root@server server]# mkdir -p /vagrant/provision/server/protect/etc/fail2ban/jail.d cp -R /etc/  
fail2ban/jail.d/customisation.local  
mkdir: invalid option -- 'R'  
Try 'mkdir --help' for more information.  
[root@server server]# mkdir -p /vagrant/provision/server/protect/etc/fail2ban/jail.d  
[root@server server]# cp -R /etc/fail2ban/jail.d/customisation.local /vagrant/provision/server/p  
rotect/etc/fail2ban/jail.d/  
[root@server server]# gedit /vagrant/provision/server  
(gedit:7569): dconf-WARNING **: 20:46:46.692: failed to commit changes to dconf: Failed to execu
```

Рис. 11: Журнал Fail2ban с ignoreip

Подготовка provisioning-конфигурации

A screenshot of a code editor window titled 'protect.sh' with the path '/vagrant/provision/server' shown below the title. The window has a light gray background and a dark gray title bar. On the left side of the title bar, there are buttons for 'Open' (with a dropdown arrow) and a file icon. On the right side, there are buttons for 'Save', a hamburger menu icon, and a close 'x' button. The main area of the window contains a shell script with 10 lines of code, each preceded by a line number. The code is as follows:

```
1 #!/bin/bash
2 echo "Provisioning script $0"
3 echo "Install needed packages"
4 dnf -y install fail2ban
5 echo "Copy configuration files"
6 cp -R /vagrant/provision/server/protect/etc/* /etc
7 restorecon -vR /etc
8 echo "Start fail2ban service"
9 systemctl enable fail2ban
10 systemctl start fail2ban
```

Рис. 12: Скрипт protect.sh

Выводы по проделанной работе

В ходе лабораторной работы была реализована базовая защита сервера от атак типа «brute force» с использованием Fail2ban. Настроена защита SSH, веб-служб и почтовых сервисов, проверены механизмы блокировки, разблокировки и исключений IP-адресов. Дополнительно выполнена автоматизация настройки Fail2ban в процессе развертывания виртуальной машины с помощью Vagrant.